



SCOPE OF ACCREDITATION TO ISO/IEC 17025:2005
& ANSI/NCSL Z540-1-1994

AMETEK CTS US
 52 Mayfield Ave
 Edison, NJ 08837
 Mary Jane Salvador Phone: 732 417 0501
 E-mail: maryjane.salvador@ametek.com
 www.teseq.com

CALIBRATION

Valid To: May 31, 2018

Certificate Number: 2273.01

In recognition of the successful completion of the A2LA evaluation process, accreditation is granted to this laboratory to perform the following calibrations¹:

I. Electrical – DC/Low Frequency

Parameter/Equipment	Range	CMC ^{2, 4} (±)	Comments
EFT/Burst Generator ³ –			
Voltage	20 V to 8 kV	2.3 %	Verification of generator conformance to the waveform parameters and levels of: IEC 61000-4-4, EN 61000-4-4, GR 1089 CORE, ANSI C37.90, ISO 7637-2 Tektronix TDS 5052B, Tektronix TDS 620B, Tektronix DPO4054, Schaffner CAS 3025
Rise Time	5 ns ± 20 %	2.9 %	
Pulse Width	(35 to 200) ns	2.4 %	
Burst Duration	(0.5 to 20) ms	0.63 %	
Burst Period	(100 to 300) ms	0.63 %	
Repetition Rate	1 kHz to 1 MHz	0.63 %	

Parameter/Equipment	Range	CMC ^{2, 4} (±)	Comments
ESD Simulators ³ –			
Tip Voltage	200 V to 30 kV ± 5 %	1.0 %	Verification of generator conformance to the waveform parameters & levels of: IEC 61000-4-2, EN 61000-4-2, IEC 801-2, ISO TR 10605, ISO 10605, SAE J1113 -13, ANSI C63.16, FORD ES-XW7T-1A278-AC, FORD EMC CS 2009, GMW 3097, DC-10614, DC-11224
Rise Time	(0.6 to 1) ns	90 ps	
Peak Current	(7.5 to 112.5) A ± 10 %	7.0 %	
30 ns Current	(4 to 60) A ± 30 %	11 %	
60 ns Current	(2 to 30) A	18 %	
65 ns Current	(2 to 20) A	9.0 %	
130 ns Current	(2 to 20) A	17 %	
180 ns Current	(2 to 20) A	11 %	
360 ns Current	(2 to 20) A	18 %	
400 ns Current	(2 to 20) A	11 %	
800 ns Current	(2 to 20) A	19 %	
RC Time Constant	600 ns ± 130 ns 300 ns ± 60 ns	35 ns 15 ns	Brandenburg HV meter 139D & 149-03 (up to 40 kV, 30 GΩ input impedance), TEK TDS 7254 with Schaffner MD102 ESD target
Transient Generator ³ (Surge Generator) –			
Front/Rise Time – Open Circuit Short Circuit	1 μs to 10 ms (1 to 100) μs	2.8 %	Verification of generator conformance to the waveform parameters and levels of: IEC 61000-4-5, EN 61000-4-5, IEC 61000-4-9, IEC 61000-4-12, ANSI C62.41, UL1449, ISO7637-2, ITU Rec K.17, K.20, K.21 GR1089CORE
Time to Half Value/Duration – Open Circuit Short Circuit	1 μs to 1000 ms 1 μs to 1 ms	2.5 %	
Open Circuit Voltage	10 V to 8 kV	2.6 %	
Short Circuit Current	1 A to 4 kA	2.4 %	
Repetition Rate	(0.1 to 100) s	0.64 %	
Ring/Oscillatory Wave Rise Time	(0.5 to 1.5) μs	2.1 %	
Ring/Oscillatory Wave Frequency	5 kHz to 1 MHz	0.64 %	
Ring/Oscillatory Wave Current	1 A to 4 kA	2.4 %	
			Tektronix TDS 5052B, Tektronix TDS 620B, Pearson model 110

Parameter/Equipment	Range	CMC ^{2, 4} (±)	Comments
PQT ³ – Output Voltage Phase Angle Pulse Rise/Fall Time	Up to 260 V (AC or DC) (0 to 359)° (1 to 5) ns	0.33 % 1.2 % 1.2 %	Verification of generator conformance to the waveform parameters & levels of: IEC 61000-4-11, EN 61000-4-11, Tektronix TDS 620B, Tektronix DPO 4054
DC Voltage – Measure ³	(10 to 100) mV (0.1 to 1.0) V (1 to 10) V (10 to 100) V (100 to 1000) V	0.13 % 0.13 % 0.13 % 0.13 % 0.13 %	HP34401A
DC Current – Measure ³	(10 to 100) mA (0.1 to 1.0) A (1.0 to 3.0) A	0.20 % 0.20 % 0.20 %	HP34401A
Resistance – Measure ³	(0 to 10) Ω (10 to 100) Ω (0.1 to 100) kΩ	1.5 % 1.5 % 1.5 %	HP34401A 4-wire measurement
AC Voltage – Measure ³ , 10 Hz to 100 kHz	(10 to 100) mV (0.1 to 1.0) V (1 to 10) V (10 to 100) V (100 to 750) V	1.1 % 1.1 % 1.1 % 1.1 % 1.1 %	HP34401A
AC Current – Measure ³ , 10 Hz to 5 kHz	(10 to 100) mA (0.1 to 1.0) A (1.0 to 3.0) A	0.86 % 0.86 % 0.86 %	HP34401A
AC Voltage – Measure ³	(1 to 100) V	2.5 %	Tektronix TDS 5052B, Tektronix TDS 620B, Tektronix DPO 4054, Schaffner MD 200
AC Current – Measure ³	(0.1 to 100) A	2.4 %	Tektronix TDS 5052B, Pearson model 110 Tek-A622

II. Electrical – Microwave/RF

Parameter/Range	Frequency	CMC ^{2, 4} ($\pm$)	Comments
Amplitude Accuracy – Measure / Compare			
(>10 to 20) dBm	10 kHz to 6000 MHz	5.2 %	Power meter
(> 0 to 10) dBm	10 kHz to 6000 MHz	4.0 %	
(-40 to 0) dBm	10 kHz to 6000 MHz	3.9 %	
CDN & CDNE –			
Common Mode Impedance	10 kHz to 300 MHz	2.2 Ω	IEC 61000-4-6, CISPR 22, CISPR 16-1-2, R&S ZVRE w/ TOSM
Voltage Division Factor	10 kHz to 300 MHz	0.80 dB	
Transmission Bandwidth	(1 to 100) MHz	0.60 dB	
Crosstalk	(1 to 100) MHz	1.6 dB	
LISN –			
Common Mode Impedance	9 kHz to 30 MHz	0.60 Ω	CISPR 16-1-2, R&S ZVRE w/ TOSM
Phase Angle	9 kHz to 30 MHz	0.20°	
Voltage Division Factor	9 kHz to 30 MHz	0.10 dB	
Isolation	9 kHz to 30 MHz	6.0 dB	
ISN –			
Common Mode Impedance	150 kHz to 230 MHz	4.0 Ω	CISPR 16-1-2, CISPR 22, CISPR 32 R&S ZVRE w/ TOSM
Phase Angle	150 kHz to 30 MHz	1.1°	
Voltage Division Factor	150 kHz to 230 MHz	0.60 dB	

Parameter/Range	Frequency	CMC ² (±)	Comments
ISN – (cont)			CISPR 16-1-2, CISPR 22, CISPR 32 R&S ZVRE w/ TOSM
Longitudinal Conversion Loss	150 kHz to 300 MHz	0.60 dB	
Isolation	150 kHz to 30 MHz	1.0 dB	
Transmission Bandwidth	150 kHz to 250 MHz	0.60 dB	
Crosstalk	(1 to 250) MHz	1.2 dB	
EM Clamp			
Common Mode Impedance	10 kHz to 100 MHz	5.3 Ω	CISPR 16-1-4, IEC/EN 61000-4-6 R&S ZVRE w/ TOSM
S Parameters –			
S ₁₁	9 kHz to 4000 MHz	2.2 dB	R&S ZVRE w/ TOSM one port device
S ₂₁	9 kHz to 4000 MHz	1.4 dB	
Phase Angle	9 kHz to 110 MHz	0.2°	R&S ZVRE w/ TOSM

III. Time & Frequency

Parameter/Equipment	Range	CMC ^{2,4} (±)	Comments
Frequency – Measure	100 Hz to 1 MHz	0.7 %	Tektronix TDS 5052B, Pearson model 110

¹ This laboratory offers commercial calibration service.

² Calibration and Measurement Capability Uncertainty (CMC) is the smallest uncertainty of measurement that a laboratory can achieve within its scope of accreditation when performing more or less routine calibrations of nearly ideal measurement standards or nearly ideal measuring equipment. CMCs represent expanded uncertainties expressed at approximately the 95 % level of confidence, usually using a coverage factor of $k = 2$. The actual measurement uncertainty of a specific calibration performed by the laboratory may be greater than the CMC due to the behavior of the customer's device and to influences from the circumstances of the specific calibration.

³ Field calibration service is available for this calibration and this laboratory meets A2LA R104 – *General Requirements: Accreditation of Field Testing and Field Calibration Laboratories* for these calibrations. Please note the actual measurement uncertainties achievable on a customer's site can normally be expected to be larger than the CMC uncertainty found on the A2LA Scope. Allowance must be made for aspects such as the environment at the place of calibration and for other possible adverse effects such as those caused by transportation of the calibration equipment. The usual allowance for the actual uncertainty introduced by the item being calibrated, (e.g. resolution) must also be considered and this, on its own, could result in the actual measurement uncertainty achievable on a customer's site being larger than the CMC.

⁴ In the statement of CMC, all percentages are stated as percent of reading unless otherwise noted.



Accredited Laboratory

A2LA has accredited

AMETEK CTS US

Edison, NJ

for technical competence in the field of

Calibration

This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2005 *General requirements for the competence of testing and calibration laboratories*. This laboratory also meets the requirements of ANSI/NCSLI Z540-1-1994 and R205 – Specific Requirements: Calibration Laboratory Accreditation Program. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (refer to joint ISO-ILAC-IAF Communiqué dated 8 January 2009).



Presented this 16th day of August 2016.

A handwritten signature in blue ink, appearing to read 'L. A. ...', written over a horizontal line.

President and CEO
For the Accreditation Council
Certificate Number 2273.01
Valid to May 31, 2018
Revised on February 9, 2017

For the calibrations to which this accreditation applies, please refer to the laboratory's Calibration Scope of Accreditation.